

Aim:

to setup and access the azure devops environment by creating an organization through the azure portal.

Installation:-

1) open your web browser and go to the azure website:

~~<https://azure.microsoft.com/en-us/get-started/azure-portal>~~

signin using your microsoft account credentials
if you don't have a microsoft account, you can create one here:

~~<https://signup.live.com/?lie=1>~~

2) in azure home page, open devops environment in azure platform by typing azure devops organization in the search bar.

3) click on my account devops organization link, and create an organization and you should be taken to the azure devops organization home page.

Result:

Successfully accessed the azure devops environment and created a new organization through the azure portal.

SETTING UP EPICS, FEATURES AND USERS STORIES FOR PROJECTS PLANNING

Aim:

To learn about creating and organizing Epic, Features and tasks in azure devops for an E-commerce web application.

Project Description

The project is an E-commerce System that allows user to browse products, add items to a cart, and make purchases online. The application includes user authentication, Product management, shopping cart functionality, order processing and notification system.

(I) Epic created:

- 1) User Authentication System
- 2) Product management System
- 3) Shopping cart system
- 4) order and payment management.
- 5) Notification and delivery system

(II) Features created

- 1) User sign-up and login
- 2) Browse and Search Product.
- 3) Add to cart.
- 4) Place orders
- 5) Order tracking

7) Agile workflow Setup:

Azure devops Supports agile methodologies by providing tools such as Kanban boards, Sprint planning and backlog management.

Team can:

- organize work into iterations
- Prioritize tasks using backlogs.
- track process visually using Kanban board
- Monitor team performance through analytics and reports.

Result:

Successfully created and configured an azure devops project environment with a structured organization, project setup and agile workflow integration.

4) Accessing and Navigating the organization

After logging into azure devops it is important to ensure that you are working within the correct organization, especially if you are part of multiple organization to verify or switch organization:

- locate the organization name in the top-left corner
- click on it to view a list of available organizations
- select the desired organization to switch context.

5) Understanding the Project dashboard

The project dashboard provides a comprehensive overview of the project's status and progress. It acts as a centralized interface where team member can monitor ongoing activities and access various tools.

Key components of the dashboard include:

- Overview Section: Display project summary and recent activity
- Boards: used for managing work items such as user stories, tasks and bugs.
- Repos: Host source code repositories and version control workflow
- Artifacts: manages packages and dependencies.

Ex no: 2

AZURE DEVOPS PROJECT SETUP AND USER

STORY MANAGEMENT

Aim:

to establish an azure devops Project environment that enables efficient team collaborating, Streamlined development work flows, and effective agile projects management through structured planning, tracking and execution of tasks.

Procedure:

1) creating an azure account:

The first step is setting up an azure devops environment is to create a microsoft Azure account server as the foundation for accessing all azure services, including azure devops

To begin, navigate to the official microsoft azure website and select the option to create a new account. You will be required to provide basic details such as your email address, phone number, and payment information. microsoft may offer a free trial credit for new users, which can be utilized during initial step and examination.

III User stories Details

1) As a user, I want to create an account and then I have to login

2) As a user, I want to browse a product and search a product by category

3) As a user, I want to add a product to cart or I can edit or update

4) As a user, I want to place an order and then to make payment

5) As a user, I want to track my order and to get notification

(iv) Task created:

>> Design login and signup UI

>> Implement authentication (login)

>> Develop product listing and search functionality

>> Create product detail pages

>> Implement shopping cart functionality

>> Develop order placement system

Result:

Thus, the creation of epic, Features, user stories and tasks for the E-commerce System has been successfully completed using DevOps

user
user ID
name
email
login()
browse Product()

Cart
Cart ID
Created Date
add Product()
remove Product()

Order
Order ID
Order Date
Total Amount
Place Order()
view Order()

Product
Product ID
Product Name
Category
Price
get Product Details()

Admin
Add Product()
delete Product()
update Product()

DESIGNING CLASS AND SEQUENCE DIAGRAM FOR PROJECT ARCHITECTURE

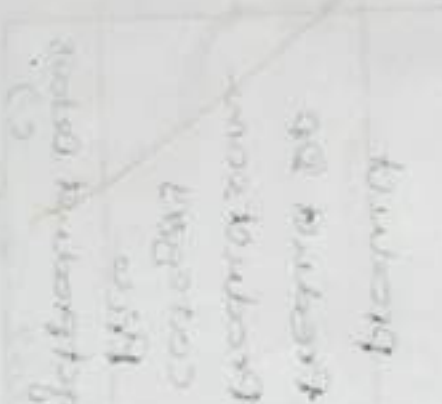
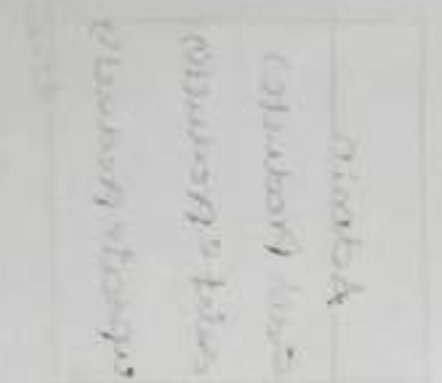
Aim:-

To design a class diagram and sequence diagram for E-commerce site to represent the structure and interaction of system components.

Observation:

The E-commerce system is used to manage products, user account, orders, and payments on an online platform.

The class diagram represents the static structure of the system. It shows the classes such as user, product, cart, order, admin and database manager, along with their attributes and methods.



Aim:

to create POKER estimation for the user stories of the E-commerce website project using azure devops.

Poker Estimation

Poker estimation is used to assign story points based on complexity, effort, and time required for each user story.

User Stories estimation:-

- » user sign-up - 3 story points
- » user login - 3
- » Browse product - 5
- » view product - 3
- » Add to cart - 5
- » update cart - 5
- » wishlist functionality - 3
- » checkout process - 3
- » order placement - 5
- » order history - 3
- » Testing and integration - 5

Result:

The story points for all user stories have been successfully assigned using POKER Estimation for the E-commerce web project.

Sprint 4: order tracking & notification.

- » Track order status
- » Display order history
- » Send email/SMS notifications
- » delivery status updates

Sprint 5: Admin & testing

- » Admin panel for product management
- » Add/edit/delete product
- » manage user order
- » Integrate all modules
- » testing and debugging of applications.

Result:

The sprint have been successfully created
and user stories have been assigned for
E-commerce project using Azure devops

Aim:

To assign user stories to specific sprints for the E-commerce system, project using Azure DevOps.

SPRINT PLANNING

Sprint 1: Authentication & user setup

- » user sign-up functionality.
- » user login functionality.
- » Basic UI design for login and signup
- » validation of user credentials

Sprint 2: Product Browsing & cart

- » Browse Product by category
- » Search Product functionality
- » view Product details
- » Add Products to cart.
- » Delivery status update.

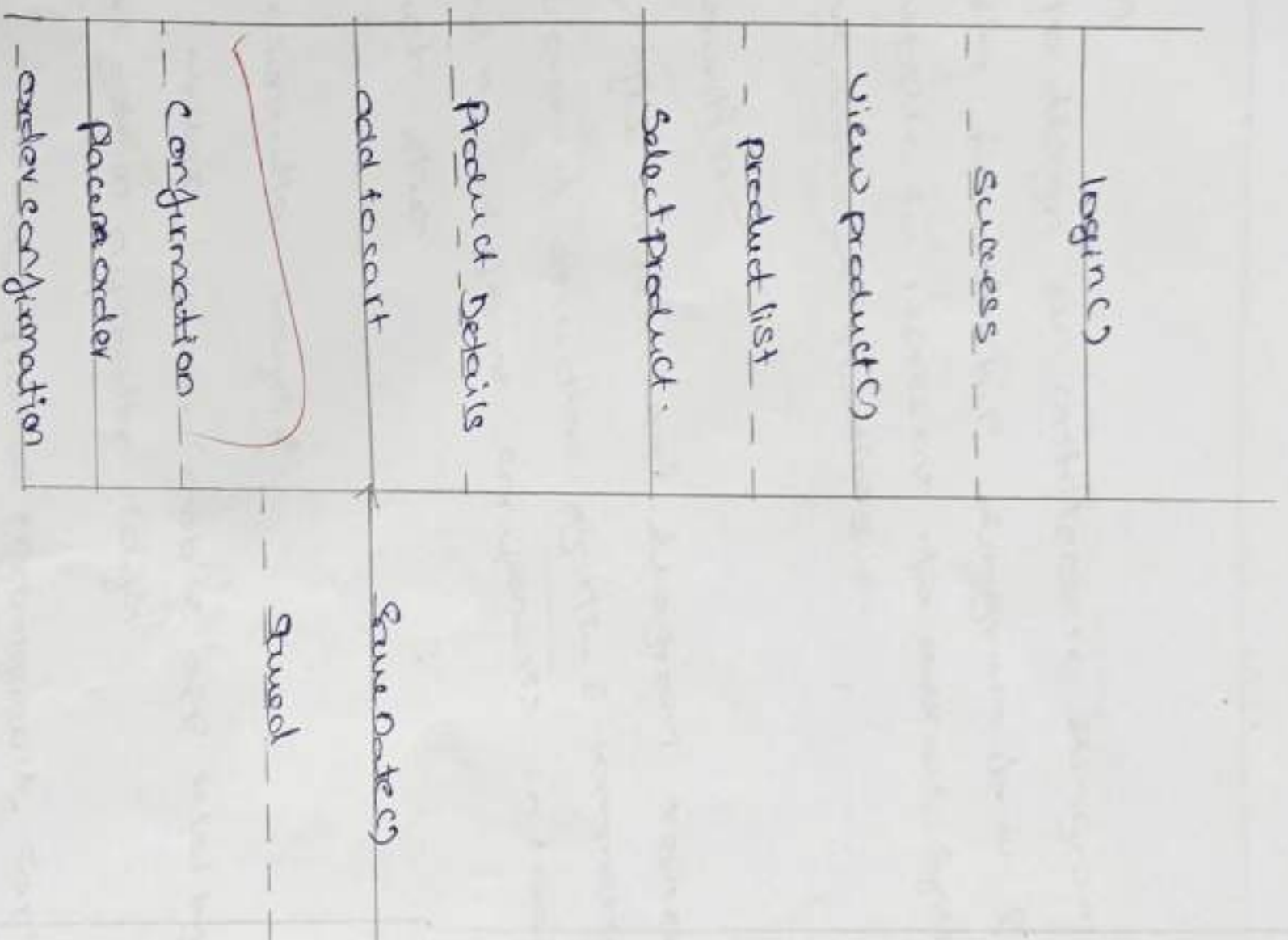
Sprint 3: Order & Payment management

- » place order functionality
- » Address and delivery details form
- » Integrate payment gateway
- » order confirmation system
- » generate invoice/receipt.

User

System

~~System~~ Database



Result

Thus the class diagram and sequence diagram ~~has been~~ for E-commerce has been successful verified.

TESTING

Aim:-

To design test plan and test cases for the E-commerce web application project.

Description:-

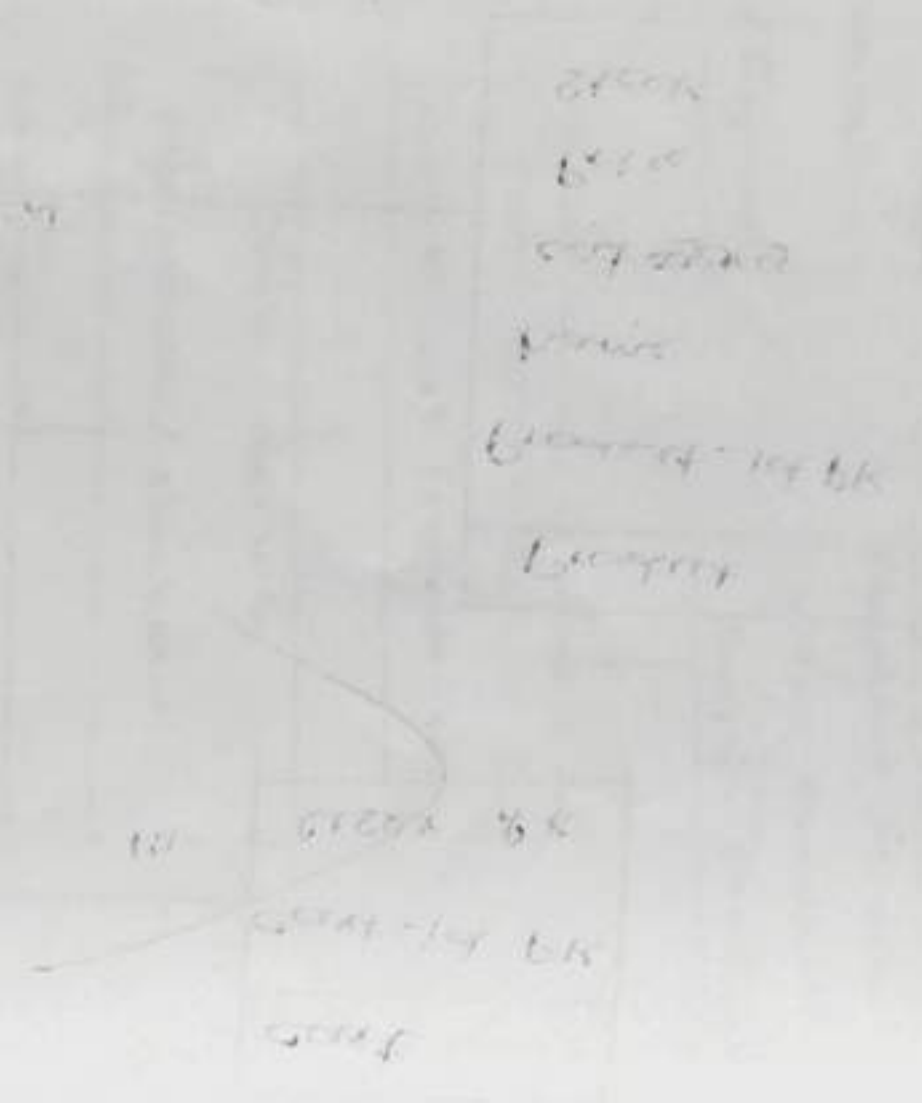
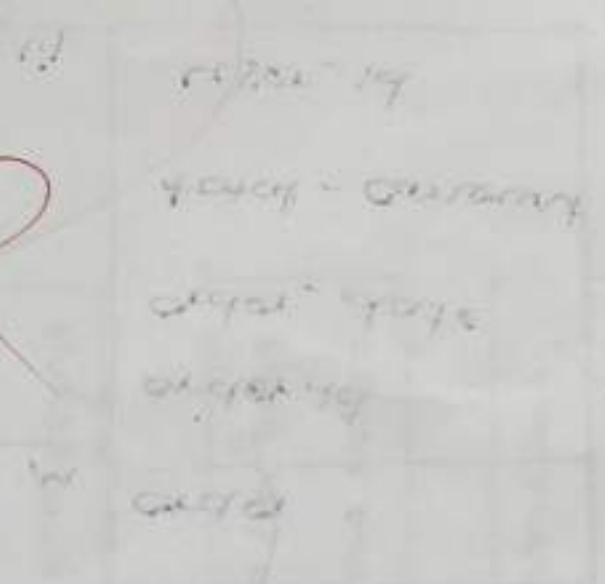
Test plan were created for key features like user registration login, product browsing, cart management, and order processing. Both happy path and error scenarios were tested to ensure system reliability and performance.

Sample Test case:-

- » Login Success: user enters valid credentials → redirected to homepage/dashboard.
- » Login Failure: Invalid username / password → error message display
- » User Registration: valid details entered → account created successfully
- » Add to cart: Product added → item appears in cart
- » Remove from cart: Product added → cart updated correctly
- » Empty cart checkout: No item in cart → error message display
- » Place order: valid address & payment → Order Place Successful
- » Invalid Payment: Payment failed → Transaction error screen

boundary between
two
systems

~~Result:~~
Thus the ER diagram for the Commerce
Site was successfully designed and
analysis



user	
user-id	PK
name	
email	
phone	
password	

cart	
cart-id	PK
user	PK

Product	
Product-id	PK
name	
category	
Price	
Stock	

order	
order-id	
order-date	
total-amount	
user-id	

Payment	
Payment-id	
amount	
payment method	

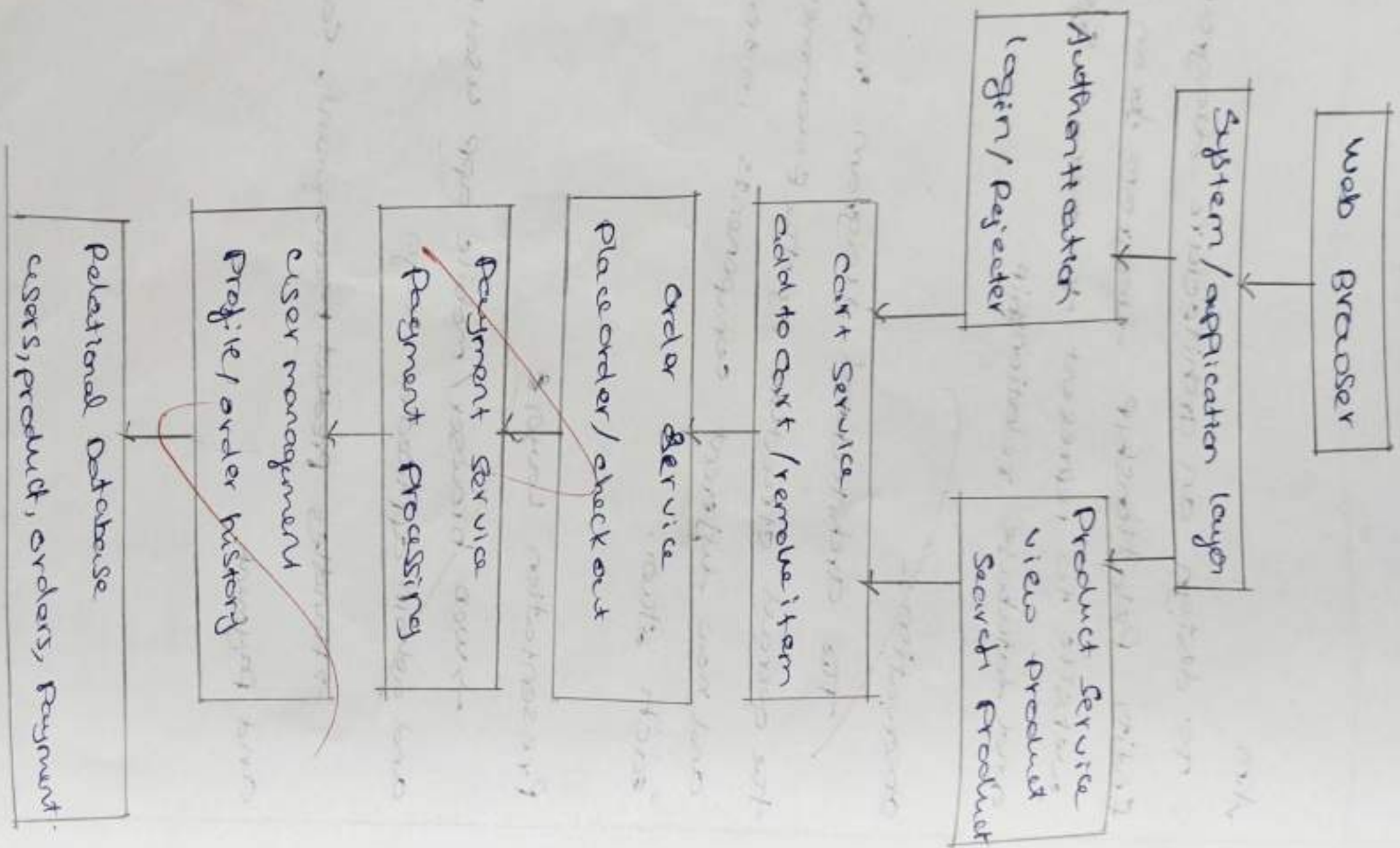
1:1

1:1

M

M:N

1:1



DESIGNING ARCHITECTURE AND OF A SYSTEM FOR PROGRAMS STRUCTURE.

Aim:

To design an architecture diagram and Entity Relationship diagram for an E-commerce website to represent the overall system and database relationship.

Observation:-

The architectural diagram represents the overall structure of the E-commerce system and how different components interact with each other.

Presentation Layer

→ Web browser / mobile app used by user and admin application layer

→ Handles Product management, Cart, orders and Payment

Database Layer

Stores product information, user details, orders, etc.

>> /static ->

- css/ -> style

- js/ -> scripts

- images/ -> product page

Result:

The e-commerce Project structure was organized clearly, improving reliability, scalability, and maintainability of the application.

Aim:-

To organize the e-commerce Project Structure and maintain proper naming convention for better Scalability and maintainability

Project Structure Example:-

- >> /Project -> main application code
- >> /templates -> HTML files
- >> /Static -> CSS, javascript, image
- >> /tests -> test case.
- >> app.py/main.py -> main application file.

Naming conventions:-

- >> files: lowercase with underscores
- >> variables: meaningful name
- >> Functions: action-based naming.

Project structure Example:-

- >> /Project -> main application code
- >> /templates ->

- home.html -> homepage

- Product.html -> Product details

- cart.html -> Shopping cart

- checkout.html -> Payment page

Load testing and CI/CD pipeline

The CI/CD pipeline for load testing and deployment was automated using Jenkins and Docker.

The load testing was performed using JMeter and the results were analyzed using Grafana and Prometheus.

- 1. Load testing was performed using JMeter.
- 2. The results were analyzed using Grafana and Prometheus.
- 3. The CI/CD pipeline was automated using Jenkins and Docker.
- 4. The deployment was performed using Ansible.

Result:-

Load testing and CI/CD pipeline for the E-commerce project were completed successfully, ensuring high performance and efficient automation.

Load Testing and Pipelines

Aim:-

To perform the load testing and implement CI/CD pipeline for the E-commerce web application.

Description:-

Load testing was conducted on the E-commerce application using Azure to evaluate performance under high user traffic. Metrics include Product Search, cart updates, and checkout request.

*

Pipeline task:

- Code checkout from Github repository
- Setup Node.js/Python environment
- Install dependencies
- Run test scripts
- Build the application
- Deploy to cloud environment.

Outcome:-

➤ The E-commerce application successfully handles multiple concurrent user performing operations like browsing, adding to cart, and placing order.

➤ CI/CD pipeline automated the build, testing, and deployment process, ensuring faster and reliable update.

View Product: Product list displayed correctly

>> out of stock Product: Added cart disabled

add new product button not visible or

button not working or error message

search bar not working or error message

product list not displayed or error message

add to cart button not working or error message

checkout button not working or error message

payment gateway not working or error message

error message

product list not displayed or error message

add to cart button not working or error message

checkout button not working or error message

error message

product list not displayed or error message

error message

add to cart button not working or error message

checkout button not working or error message

error message

Result:-

Test case were successfully created

and executed for major user function

ensuring proper workout of the E-commerce

application

6) Managing user stories using agile boards

User stories are a fundamental component of agile method. They represent features or requirements for the end-user perspective and help items plan and prioritize work effectively.

To manage user stories in azure DevOps:

- From the left-hand navigation menu,

Select "Board"

This will open the main board interface where you can view backlog, sprints, and work items

To create a new user story:

- Locate the option labeled "Add a work

item at the top of the page

Alternatively, use the "+" icon or add new work item" option depending on the current view

- From the dropdown menu, select "User stories"

A form will appear where you can enter details information including:

- Title: A concise description of the user requirement

- Description: A detailed expansion of the feature or functionality

- Acceptance criteria: condition that must be met for the story to be considered completely

- Assigned to: Specifies the team member ^{site} responsible for the task